terraform_cheatsheet

Terraform describes infrastructure as code: init downloads providers, plan shows what will change, apply makes it so, and the state file remembers what exists. These are the commands you run every day, grouped by task. OpenTofu accepts the same commands: switch to tofu or a tf alias and every command updates. Click a command to copy it.

:workflow — 7 cmds

Init, plan, apply, destroy.

Download providers and set up the backend

```
$ tofu init
```

Show what will change

```
$ tofu plan
```

Apply the changes

```
$ tofu apply
```

Apply without the confirmation prompt

```
$ tofu apply -auto-approve
```

Save a plan, then apply exactly that plan

```
$ tofu plan -out=tfplan   $ tofu apply tfplan
```

Preview a destroy

```
$ tofu plan -destroy
```

Destroy everything in the configuration

```
$ tofu destroy
```

:plan — 8 cmds

Variables, targets and replacements.

Set a variable

```
$ tofu plan -var='region=us-east-1'
```

Use a variables file

```
$ tofu plan -var-file=prod.tfvars
```

Plan only one resource and its dependencies

```
$ tofu plan -target=aws_instance.web
```

Force a resource to be recreated

```
$ tofu apply -replace=aws_instance.web
```

Sync state with real infrastructure only

```
$ tofu apply -refresh-only
```

Exit code 2 when there are changes (for CI)

```
$ tofu plan -detailed-exitcode
```

Wait for the state lock

```
$ tofu apply -lock-timeout=5m
```

Run more operations at once

```
$ tofu apply -parallelism=20
```

:init — 8 cmds

Backends, provider versions and modules.

Upgrade providers and modules within their constraints

```
$ tofu init -upgrade
```

Reconfigure the backend, ignoring the old one

```
$ tofu init -reconfigure
```

Move state to a new backend

```
$ tofu init -migrate-state
```

Pass backend settings from a file

```
$ tofu init -backend-config=backend.hcl
```

List the providers in use

```
$ tofu providers
```

Lock provider hashes for several platforms

```
$ tofu providers lock -platform=linux_amd64 -
platform=darwin_arm64
```

Update modules

```
$ tofu get -update
```

Show the CLI and provider versions

```
$ tofu version
```

:state — 8 cmds

What Terraform thinks exists.

List resources in state

```
$ tofu state list
```

Show one resource

```
$ tofu state show aws_instance.web
```

Show the whole state or a saved plan

```
$ tofu show   $ tofu show -json tfplan
```

Rename or move a resource

```
$ tofu state mv aws_instance.web aws_instance.app
```

Move a resource into a module

```
$ tofu state mv aws_instance.web
module.app.aws_instance.web
```

Stop managing a resource without destroying it

```
$ tofu state rm aws_instance.web
```

Download the remote state

```
$ tofu state pull > terraform.tfstate
```

Release a stuck lock

```
$ tofu force-unlock LOCK_ID
```

:import

3 cmds

Bring existing infrastructure under Terraform.

Import a resource into state

```
$ tofu import aws_instance.web i-0abc123def456
```

Write config for import blocks

```
$ tofu plan -generate-config-out=generated.tf
```

Replace a provider source in state

```
$ tofu state replace-provider hashicorp/aws registry.example.com/acme/aws
```

:workspaces

5 cmds

Several states for one configuration.

List workspaces

```
$ tofu workspace list
```

Show the current workspace

```
$ tofu workspace show
```

Create a workspace

```
$ tofu workspace new staging
```

Switch workspace

```
$ tofu workspace select staging
```

Delete a workspace

```
$ tofu workspace delete staging
```

:outputs

5 cmds

Read values back out.

Show all outputs

```
$ tofu output
```

Print one output without quotes

```
$ tofu output -raw db_endpoint
```

Outputs as JSON

```
$ tofu output -json
```

Try expressions against the state

```
$ tofu console
```

Draw the dependency graph

```
$ tofu graph | dot -Tsvg > graph.svg
```

:code

5 cmds

Keep the configuration clean and correct.

Format files in the current folder

```
$ tofu fmt
```

Check formatting everywhere (for CI)

```
$ tofu fmt -recursive -check
```

Validate the configuration

```
$ tofu validate
```

Run .tftest.hcl tests

```
$ tofu test
```

Log in to HCP Terraform

```
$ tofu login
```

:debug

5 cmds

Environment variables that change how the CLI behaves.

Debug logs

```
$ TF_LOG=DEBUG tofu plan
```

Trace logs to a file

```
$ TF_LOG=TRACE TF_LOG_PATH=./tf.log tofu apply
```

Set a variable from the environment

```
$ TF_VAR_region=us-east-1 terraform plan
```

Pick the workspace from the environment

```
$ TF_WORKSPACE=staging tofu plan
```

Run in another folder

```
$ tofu -chdir=envs/prod plan
```

Updated Nov 8, 2026 · checked against Terraform 1.x and OpenTofu 1.x · sources: Terraform CLI commands, OpenTofu CLI commands

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